



Webly supervised 3D shape recognition

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ABSTRACT

Recently, deep learning has become the most promising method for 3D shape recognition. However, it is expensive to obtain large-scale accurately labeled 3D datasets required for deep learning. To reduce the cost, this paper presents a study on learning 3D recognition models from 3D noisy web data. To realize the method, we crawl 3D data from the Internet to construct a new 3D shape dataset called 3DWebNet40, and propose a novel webly supervised 3D shape recognition method handling label noise and background noise simultaneously. To alleviate the downside of background noise, we utilize the region-attention module to enhance the representative local feature of the view. For label noise, we combine label selection and feature selection to jointly select training data, and train the 3D shape classifier on the selected subsets with different regularization strategies to reduce the interference of label noise. Extensive experiments on 3DWebNet40 demonstrate the effectiveness of our method¹.

1. Introduction

Deep learning techniques have made great progress in 3D shape recognition [1]. The success of these advances is due to the construction of many 3D shape datasets such as ModelNet [2] and Part-Net [3]. However, it is very labor intensive and time consuming to obtain such a carefully annotated data set. Moreover, existing 3D datasets only contain a limited number of semantic concepts, and the amount of clean training data may not be sufficient for practical needs, so deep learning is usually an expensive technique choice when faced with new recognition tasks. This prevents deep learning from rapidly promoting the development of 3D shape recognition techniques.

Normally, the process of constructing a 3D shape benchmark dataset [4] consists of two steps: data collection and annotation. Yağ et al. [5] used PlantVillage, a dataset collected by manual methods such as directly photographing plant leaves and manually annotated, to achieve real-time classification of plant diseases such as apple. Sezer et al. [6] used a high-resolution camera to take images of various printed circuit board (PCB) defect types to construct a dataset, and realized the detection and classification of solder paste defects on PCB. The rapid development of online 3D shape repositories allows us to crawl a large amount of 3D data from the Internet automatically and efficiently. In contrast, data annotation is much more cumbersome because it requires a lot of manpower and resources. Inspired by FoPro [7] and Luo et al. [8], we consider training 3D recognition depth models directly

from crawled web data, which is becoming increasingly popular in image recognition. FoPro [7] proposes a few shot-guided prototypical representation learning method which is guided by a few images from the real world domain to learn robust and generalizable representations from web data. Luo et al. [8] utilize large amounts of web data to learn a highly salient object detection model, achieving high performance by introducing a novel quality evaluation method and a self-training approach.

The idea of learning from web data is generically called web-supervised learning. The web data are crawled on the Internet using category semantics and automatically labeled without any manual annotation cost. However, webly supervised learning is challenging due to its poor quality which could interfere with the training of deep networks.

This paper proposes a study on web-supervised 3D shape recognition to alleviate the aforementioned challenges. To compare the performance of different algorithms, we use 40 semantic categories from ModelNet40 [2] to build a new 3D web shape dataset called 3DWebNet40 which contains 36,665 3D shapes crawled from the online repositories and about three times the number of shapes (i.e., 12,311) in ModelNet40. The 3DWebNet40 construction process is fully automated and does not require human involvement. Through analysis, we found that 3DWebNet40 has two types of noise. The first type of noise is label noise, which refers to incorrectly labeled data. 3D Web data

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¹ Code and datasets are publicly available online at <https://github.com/yxizhong>.

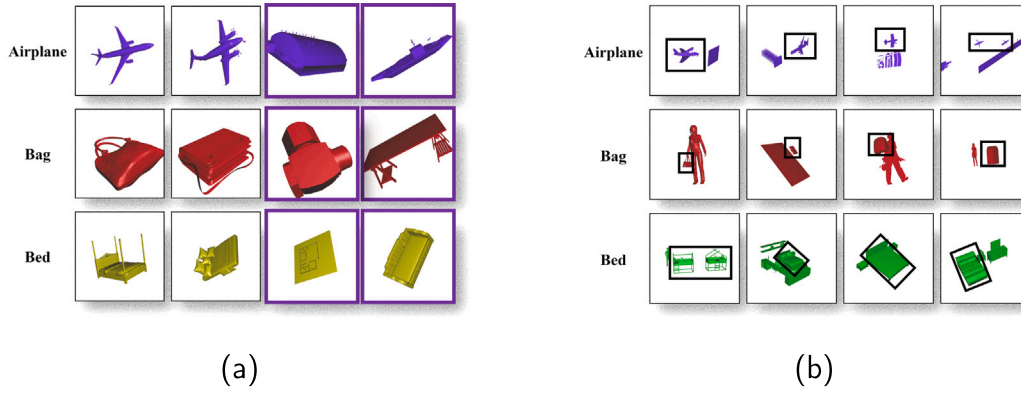


Fig. 1. Examples of 3DWebNet40: (a) Label noise (purple bounding box). (b) Background noise (the key areas of the 3D target object are outlined in the black bounding box, and everything outside this region is background noise). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

are usually labeled without any quality control, resulting in inevitable label noise in our crawled data. Fig. 1(a) shows some label noise in purple bounding boxes. One of the most popular choices to tackle label noise is sample selection which uses some strategy to separate the original dataset into clean and noisy data. However, the existing sample selection method focuses mainly on 2D images that are not directly applicable to 3D data. Therefore, selecting label noise samples in 3D shape recognition tasks is a challenging problem to be studied. The second one is background noise which means some 3D shapes are shown as complex scenes and the target object is sometimes inapparent. And the background objects usually cannot provide effective discriminative features. As shown in Fig. 1(b), the black box shows the target shape and the others are background noise. That is very different from existing standard datasets (such as ModelNet) in which the shapes are all shown as isolated objects. A straightforward approach is to generate bounding boxes to extract target geometry [9], but this also brings a lot of annotation burden.

In this work, we propose an effective webly supervised 3D shape recognition method to solve both background noise and label noise in 3D web data. Currently, there are various representations of 3D data available for 3D feature learning [1] and we chose the multiview-based approach among the existing representation methods, which represents 3D data by a set of 2D images taken from multiple views [10,11]. The method first extracts the features of each rendered image through a CNN architecture pre-trained in a massive image database and then fine-tunes on the 3D data by aggregating the multi-view features. Benefiting from the pre-training process, we believe that the multi-view 3D data can lead to a more robust model [12] for the webly supervised training than point clouds [13] and grids [14]. However, the task of recognizing 3D web data is still very challenging. Background noise makes the target object occluded or even invisible in some view projections and the label noise information could seriously interfere with the extraction of discriminative features in the process of fine-tuning.

To solve the above problem, we use the attention mechanism and sample selection strategy to deal with background noise and label noise, respectively. We initially incorporate a region attention module into the extraction of features in view to reduce the influence of irrelevant objects through the attention mechanism [15] to achieve the purpose of suppressing background noise. For label noise, we use the sample selection method, which is mainly performed in two ways: label selection and feature selection. Deep neural networks always prioritize fitting clean data [16] and label selection uses this characteristic to filter data in the early stage of network training. However, the representation of 3D models is richer than 2D images, so during early training, even clean data may prove challenging to fit. Feature selection refers to using feature decomposition to generate the principal components of the latent representation, and then filtering based on the similarity between principal components and the data. However, from a structural

point of view, 2D images are represented by regular 2D pixel matrices, while 3D models are often represented by irregular grids, which makes 3D models more complex than 2D images, resulting in some classes of principal components that may not be accurate enough [1,2]. So it is difficult to achieve satisfactory screening results with only one single selection criterion. Inspired by PES [17] and FINE [18], we propose a joint sample selection mechanism based on both label selection and feature selection to divide the training data into three categories: clean data, edge data, and hard data. The clean data are defined as the ones that satisfy both the label selection and the feature selection, the edge data are the data that meet only one of the selection criteria, and the hard data are those that do not satisfy both of the selection criteria. It is reasonable to assume that the label noise in the three types of data increases sequentially. Based on label credibility, we employ regularization methods with varying degrees of noise resistance in the three subsets for training. Clean data is trained using standard cross-entropy loss and edge data utilize the generalized cross-entropy (GCE) loss [19]. Then, we merge the hard data with the other two subsets and perform the Mixup [20] method for data augmentation, followed by loss calculation using binary cross-entropy loss (BCE). The different regularization strategies effectively and comprehensively exploit the 3D training data to enhance the robustness of the CNN model. We conducted extensive experiments on various baseline methods, benchmarking the proposed 3DWebNet40 and the proposed method, and the experimental results demonstrate the superiority of our method over other methods. Our main contributions are described as follows:

First, we construct a 3D dataset 3DWebNet40, for fair benchmarking of webly supervised 3D shape recognition methods.

Second, we propose an efficient web-supervised 3D shape recognition method. Based on the attention mechanism and sample selection mechanism, the method can handle both background noise and label noise and does not require additional manual annotation.

Finally, we combine label selection with feature selection to jointly select 3D label noise, which further improves the accuracy of principal component decomposition under feature representation and reduces the bias of label confidence prediction.

2. Related work

This work relates to several research areas. In this section, we mainly focus on 3D shape datasets, 3D shape recognition, webly supervised learning, and label noise learning.

2.1. 3D shape dataset

In recent years, a large number of datasets have been built to evaluate 3D shape retrieval and recognition algorithms. Currently, the two largest 3D shape benchmark datasets are ShapeNetCore55 [4]

Table 1
Comparison of 3D shape datasets.

Dataset	Size	Classes	Contain	Time
ShapeNetCore55	51,300	55	3D shape	2015
PartNet	26,671	24	3D shape	2019
ModelNet40	24,000	40	3D shape	2021
ABO	8000	63	Real images, 3D shape, PBR	2022
3DWebNet40	36,665	40	3D shape	2024

and ModelNet40 [2]. ShapeNetCore55 contains more than 50,000 3D shapes and 55 categories, while ModelNet40 contains more than 20,000 shapes and 40 categories. In addition, PartNet [3] is a large-scale consistent data set with 26,671 3D shapes covering 24 object categories. And there are also some photo-realistic datasets mainly applied in the field of 3D shape reconstruction, such as ABO [21]. We briefly summarize and compare the above data sets in Table 1.

The above datasets are capable of supporting deep learning applications. However, the production of these 3D object datasets is time-consuming and labor-intensive, requiring a large amount of human effort to annotate the datasets. In contrast, our dataset is obtained directly from the Internet without interactive cleaning and annotation, thus significantly reducing time and labor costs. In addition, we obtained a high-quality test set through manual screening after data collection, and evaluated the noise ratio of the training set (Section 3) to verify the correctness of the data set. Since there is no manual annotation, our dataset contains a lot of noise data, which poses many challenges for the implementation of deep learning. To the best of our knowledge, this is the first 3D shape benchmark dataset for a webly supervised 3D shape recognition task.

2.2. 3D shape recognition

The 3D shape recognition method usually consists of two steps: feature extraction and classifier training. The feature extraction step obtains a global feature vector from the raw representation of the 3D shape, while the classifier training step learns a classifier on the given labeled 3D shape dataset. At the early stage, feature extraction and classifier learning are two separate steps. Specifically, the feature extraction step does not require any labeled 3D data and directly obtains the feature representation from the unlabeled data using unsupervised learning by minimizing the reconstruction loss [22]. However, since the classifier learning must rely on the label information of the data, the classification performance of the above methods is naturally unsatisfactory. Currently, 3D recognition tasks are mainly divided into two ways: supervised learning and semi-supervised learning.

Recently, supervised methods [13,23] have become the dominant technique for 3D object recognition. They learn from different 3D primitive representations (e.g., point clouds [13] and views [10,11]) to learn feature representation and train classifier in an end-to-end manner via deep learning. Point-TnT [13] is a two-stage point feature learning approach that combines local and global attention modules. Supervised learning methods typically show state-of-the-art performance in 3D shape recognition, but it is worth noting that the above methods require a large number of expensive labeled data to learn the classifier, whether the feature extraction step is supervised or unsupervised.

To reduce the burden of annotation, an effective approach is semi-supervised learning [24], which can utilize both labeled and unlabeled data with pseudo-labeling or consistency constraint for 3D shape recognition. Song et al. [24] use a co-training algorithm to achieve end-to-end training by pseudo-labeling unlabeled samples. However, to ensure the performance of semi-supervised learning, the labels of the labeled data should be clean and the unlabeled data should not contain any irrelevant objects. In contrast, our 3D shape recognition method does not require high-quality labeled data so the construction cost of the training dataset is lower compared to the other methods, and it has better performance for complex 3D web shape recognition.

2.3. Webly supervised learning

Web data have attracted a great deal of research interest in recent years, especially in the field of the image. There have been many studies on using web images to learn classifiers [25], and most works only focus on the label noise. In addition, while handling label noise there are some recent works dealing with image background noise simultaneously by attention graph [25], object localization [26], and multi-instance learning [27].

For example, Zhuang et al. [25] propose a weakly supervised framework for learning visual representations from 2D Web data, which is trained by randomly grouping images and localizes discriminative regions from correctly labeled images in the combined features by a spatial attention mechanism. However, the spatial attention mechanism leads to the loss of channel information and also ignores the inter-dimensional dependencies. In contrast, the attention mechanism in our work uses a multi-branch structure to capture cross-dimensional interactions to calculate attention weights, which can obtain a richer feature representation and assist us in dealing with the background noise problem more accurately. Moreover, these works are all centered around simulated noisy data (e.g., CIFAR100 [28]) or 2D real datasets (e.g., WebVision [29]) while the method proposed in this work is trained based on 3D web data and handles 3D background noise and 3D label noise at the same time.

2.4. Label noise learning

A large amount of work currently exists on label noise and the main existing strategies to deal with label noise are robust loss functions [30,31], loss adjustment [32], and sample selection [17,33]. Robust loss function refers to appropriately modifying the loss function to make the loss of the clean data smaller than noisy data during the training process. However, it is only suitable for simple training scenarios and is difficult to achieve good performance when applied alone to the learning of 3D web data. Loss adjustment is the process of adjusting the loss according to the confidence of a given loss(or label) by loss correction, loss reweighting, etc. However, it usually relies on prior knowledge of the dataset to accurately estimate the transition matrix. Sample selection means identifying true labeled examples from noisy training data, and the diversity and extension space of selecting methods make it more suitable for dealing with noisy data in complex datasets. The selection criteria and timing of selection are particularly important for accurate screening results. In this paper, we extend the sample selection strategy to 3D shapes. However, the 3D model representation is more complex compared to 2D images and we cannot make an accurate judgment on the timing of sample selection in the training process when the noise proportion of the dataset is unknown. Therefore, unlike the above sample selection methods using a single criterion (e.g., loss, predicted label, and feature representation), we perform a joint selection of the 3D web label noise by combining label selection and feature selection to obtain more reliable screening results. Inspired by DISC [34], we apply different regularization strategies for the subsets obtained through sample selection to network training. Different from DISC, we jointly filter the data through label selection and feature selection, and we do not discard the selected subset with the lowest confidence instead of using the Mixup method [20] for model training to maximize the role of all training data.

3. Dataset construction

In this section, we explain the construction details of our 3DWebNet40 dataset, which was created to evaluate the performance of web-supervised 3D shape recognition. All 3D shapes in our dataset are collected from online repositories by providing several keywords as semantic categories. Unlike previous datasets such as ModelNet [2] or

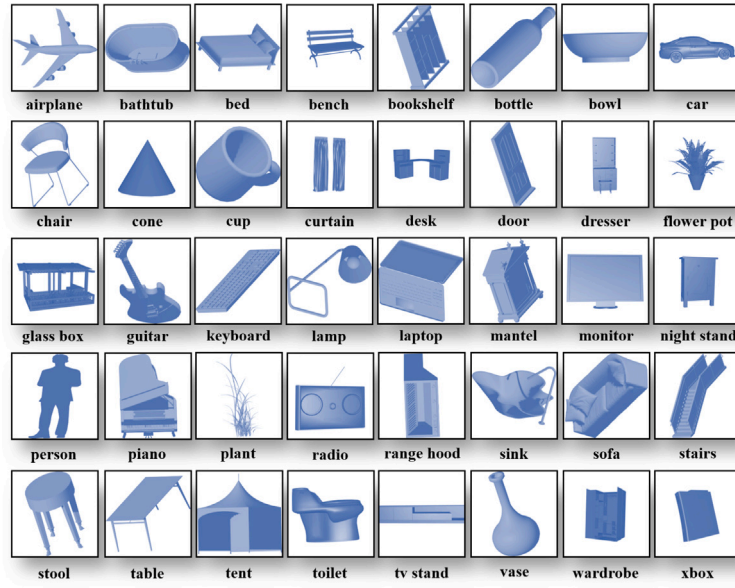


Fig. 2. The categories and the representative shape in 3DWebNet40.

ShapeNet [4], our constructing process does not contain any interactive annotation, and the category of a 3D shape is labeled with the corresponding keyword. Thus, 3DWebNet40 contains a large amount of noisy information, since the 3D shapes in the online repositories are usually casually annotated by their uploaders.

Categories: The first problem in creating the benchmark dataset is to determine the categories of 3D shapes that we will collect from the web. Specifically, we follow the semantic categories of ModelNet40 to create 3DWebNet40 and ModelNet40 consists of 12,311 shapes classified into 40 categories. The taxonomy of our 3DWebNet40 is shown in Fig. 2.

Data collection: We chose 3D Warehouse as our web resource because it is larger, more diverse, and of higher quality than other existing online 3D shape repositories. We used the crawler tool to capture the 3D model data by using the corresponding synonym list as the query keyword. Specifically, we used the category concept of ModelNet40 as the index item to capture the first 800 shapes in 3D Warehouse to compose the 3DWebNet40 data set. During the crawl process, we delete data where the model is empty or there are problems such as fractures. After the data collection is completed, we do not manually annotate the data set like ModelNet40 and ShapeNetCore55, but directly use the keywords of the index as the markup of the three-dimensional data of the Internet. The entire acquisition process took less than a day, which was performed automatically by the program without too much human involvement, and we ended up with nearly 30,000 3D shape files, more than 50G in size.

Data rendering and division: During the capture process, we convert the skp format to obj format, which is represented by the polygon surface geometry. Based on the obj format, we use blend software to render 12 projection views of each 3D shape based on the blender code provided by MVCNN [10]. Among them, 12 views are from 12 viewpoints evenly distributed at 30° intervals on the plane raised 30° above the horizontal plane of the 3D model. In this process, we directly remove the 3D shapes that cannot be converted to obj file format or cannot be rendered from the dataset. Finally, we obtain 3DWebNet40 containing 36,665 3D shapes. Note that in the specific use process, it needs to be normalized by methods such as transforms. After crawling the data, we select clean shapes for each class in the crawled 3DWebNet40 dataset as the testing set and the remaining data as the training set. During the selection process, we balance the number of training and test sets for each class. Finally, after manual selection and joint confirmation by several persons, a total of 6854 clean 3D

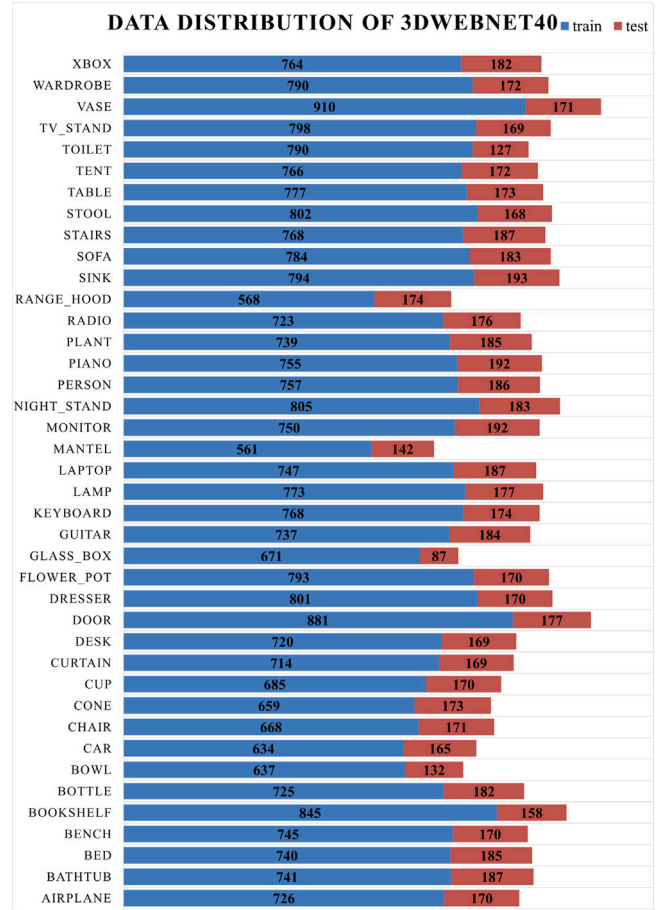


Fig. 3. The plots of the distribution of 3DWebNet40.

shapes are chosen as our 3DWebNet40 test data, and 29,811 3D shapes are included in the training set. Fig. 3 shows the distribution of the number of shapes per category for 3DWebNet40, including the number of training and testing samples.

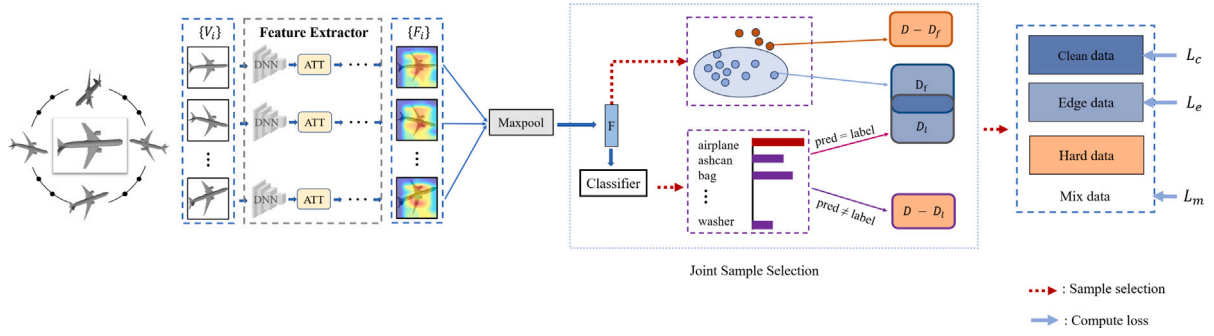


Fig. 4. Our overall structure. In the results of the joint sample selection, the intersection of D_f and D_l is clean data in dark blue, in light blue is edge data, and the concatenation of the two orange data sets is hard data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Accuracy of training data estimated for 3DWebNet40. BN and LN indicates the data with background noise, and label noise respectively.

Dataset	Clean (%)	BN (%)	LN (%)
3DWebNet40	62.2	16.6	21.2

Accuracy of the dataset: As mentioned earlier, there are two types of noise in 3DWebNet40, label noise and background noise. To estimate the difficulty in training a deep model from these two datasets, we must determine the accuracy of the network data. We directly traverse this dataset to estimate its accuracy. Finally, rough estimates of accuracy for the data are shown in Table 2, including the proportions of clean data, data with background noise, and data with label noise.

4. Method

4.1. Overview

In this work, we propose a web-supervised 3D shape recognition approach following the mainstream multiview learning framework [11] to implement training of the 3D shape recognition model in an end-to-end manner. The 3D model classification method based on multiple views takes the 2D view of the 3D object as a set of input data, and obtains the 3D shape descriptor through a series of operations such as feature extraction and fusion of the 2D view, and finally inputs it into the classifier to get the prediction result. The core of our method is how to suppress the effect of label noise and background noise in the data set D .

The overall structure of the method is shown in Fig. 4. Firstly, we define M camera settings to render 3D shapes ($M = 12$), which is the same as the MVCNN method [10]. For each 3D training shape S , we generate M views V_i and $i \in [1, M]$, where V_i represents the i th view image. Then for each image V_i , we extract the feature vector F_i by a shared feature extraction module added with the region attention module, reducing the impact of background noise and obtaining more representative image features. Then we achieve feature aggregation for multiple views by simple max pooling to obtain the feature F , which is passed through the classification layer to obtain the prediction label. Subsequently, we implement joint sample selection with the help of sample features and label prediction results of the 3D models. Finally, we use different regularization strategies to apply the selected subsets to training. Since the multi-view capture and feature extraction are the same as the MVCNN [10], we describe the region attention feature extraction and sample selection strategy in detail in this section.

4.2. The region attention feature extraction

It is observed that when given M rendered views of a 3D shape S with background noise, the target object in some views may be completely obscured by other background objects, and even if the target

object is visible in a particular view V_i , there may be a lot of irrelevant information around the target object that could interfere with feature learning. Therefore, we introduce a region attention mechanism in the original backbone network ResNet18 [35] for view feature extraction to improve the quality of feature representation by reducing the weight of irrelevant local features in the views.

Inspired by Triplet-attention [36], we combine global average pooling and global maximum pooling to compute spatial attention and channel attention. Then we can capture the correlation between different dimensions when computing attention weights to provide a rich feature representation. Following the approach of Triplet-attention where the attention module is inserted after each bottleneck block of the backbone network, we also aim to integrate our region attention module across various layers of ResNet18. The performance of the algorithm could be improved through the stacked region attention modules. For performance reasons, we finally decided to apply the attention mechanism at the last basic block of each Stage block of the Resnet18 (please see Section 5.4 for a detailed experimental description), and the structure of the feature extraction network is illustrated in Fig. 5, including the detail of the region attention module.

When we combine the region attention module with the backbone network, given the output $F_c \in R^{C \times H \times W}$ of the previous network layer, the output $F_a \in R^{C \times H \times W}$ with the same dimension, it is sent to the next Stage for further feature extraction until the final feature representation is obtained.

In the attention module, F_c is rotated by dimension to obtain $F_h \in R^{H \times C \times W}$ and $F_w \in R^{W \times H \times C}$, and $F_c^p \in R^{2 \times H \times W}$, $F_h^p \in R^{2 \times C \times W}$ and $F_w^p \in R^{2 \times H \times C}$ are obtained by global average pooling and global maximum pooling for F_c , F_h , and F_w in dimension 0, respectively. Then we feed the matrices F_c^p , F_h^p and F_w^p into the standard convolution layer with a convolution kernel of 7, and use the sigmoid activation layer to generate the normalized attention matrices $F_c^a \in R^{1 \times H \times W}$, $F_h^a \in R^{1 \times C \times W}$ and $F_w^a \in R^{1 \times H \times C}$. Finally, the obtained weight matrices are applied to the corresponding inputs (i.e., rotated after doing broadcast element-wise multiplication), and the attention-adjusted three feature tensor $F'_c \in R^{C \times H \times W}$, $F'_h \in R^{C \times H \times W}$ and $F'_w \in R^{C \times H \times W}$ are simply averaged and aggregated by the following equation, where the overline means the 90° clockwise rotation to retain the original input shape of ($C \times H \times W$):

$$F_a = \frac{1}{3}(\overline{F_h} * F_h^a + F_c * F_c^a + \overline{F_w} * F_w^a) = \frac{1}{3}(F'_c + F'_h + F'_w) \quad (1)$$

The Triplet-Attention module enables the model to understand the image content more comprehensively by applying the attention mechanism at different scales and perspectives. And the combination of spatial attention and channel attention enables the model to identify and emphasize the features related to the target object. In addition, by learning feature representations in different directions, the model is more robust to target object recognition and less susceptible to

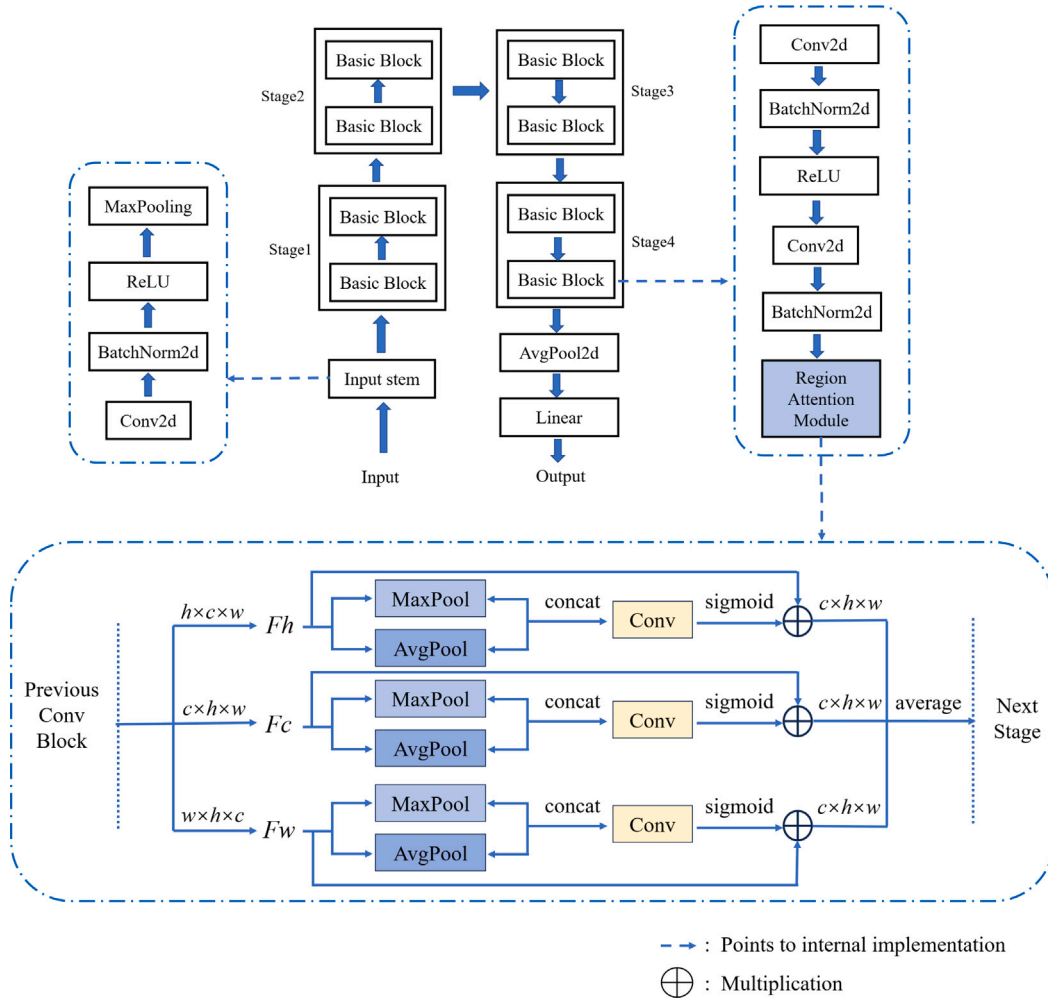


Fig. 5. The feature extraction structure with region attention module.

background changes. Therefore, its application in the 2D view feature extraction stage of 3D model classification task can be a good interference of background noise on feature representation.

4.3. Joint sample selection

The main reason for the disruption of recognition algorithm performance by label noise is that deep networks have a huge model capacity and can easily memorize and eventually overfit to noisy labels [16]. To address the characteristics of the unknown percentage, diversity, and irregularity of 3D web label noise, we choose the sample selection mechanism to mitigate the interference of label noise on algorithm learning. Most of the current standards for sample selection methods are applied to 2D images, and the selection criteria are mostly the predicted label or the sample feature. However, it is difficult to get satisfactory results using simple selection criteria because the 3D models contain richer information than the images. Therefore, we use sample features and predicted labels as the selection criteria for joint sample selection, and divide the original dataset into different subsets according to the label noise presence probability. Different regularization methods are used for processing to achieve the purpose of effectively using clean data, excluding label noise data from edge data sets and extracting clean data from hard data sets. Below we elaborate on the specific method of joint sample selection and sample division, as well as the calculation of loss value during training.

4.3.1. Details of sample selection

For joint sample selection, we select a subset D_l in the 3D training set whose predicted label l_i is the same as the given label y_i ($i \in N$, N denotes the number of 3D samples in the training set), and a subset D_f whose sample feature f_i is consistent with the potential distribution of the given label category E_{y_i} (where $y_i \in K$, K denotes the number of classes in the dataset). Algorithm 1 summarizes our sample selection algorithm.

Algorithm 1 Algorithm for Sample Selection

Require:

Original training data D , feature extractor g , classifier c , threshold for feature selection ζ , number of classes K , number of render views N

Ensure:

Clean set, D_c , Edge set D_e and Hard set D_h

- 1: Initialize $D_c \leftarrow \emptyset$, $D_e \leftarrow \emptyset$, $D_h \leftarrow \emptyset$, $G_k \leftarrow 0$ for all $k = 1, \dots, K$;
- 2: for $((x_i^1, \dots, x_i^M), y_i) \in D$ do
- 3: $(x_i^{1'}, \dots, x_i^{M'}), (x_i^{1''}, \dots, x_i^{M''}) \leftarrow$ generate two different augmentations for $(x_i^1, \dots, x_i^M)$;
- 4: $f_i \leftarrow g((x_i^1, \dots, x_i^M))$;
- 5: $pred_i' \leftarrow c((x_i^{1'}, \dots, x_i^{M'}))$;
- 6: $pred_i'' \leftarrow c((x_i^{1''}, \dots, x_i^{M''}))$;
- 7: $l_i \leftarrow \text{argmax}((pred_i' + pred_i'')/2)$;
- 8: update gram matrix $G_k \leftarrow G_k + f_i f_i^T$;
- 9: end for
- 10: $D_l \leftarrow \{((x_i^1, \dots, x_i^M), y_i) \mid l_i = y_i\}$;
- 11: for $k = 1, \dots, K$ do

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12:  $E_k \leftarrow$  the eigenvector with the largest eigenvalue of eigen decomposition of  $G_k$ ;
13: end for
14: for  $((x_i^1, \dots, x_i^M), y_i) \in D$  do
15:   Compute data alignment score  $s_i \leftarrow (E_{y_i}, f_i)^2$ ;
16:   Update the set of scores for class  $y_i$ :  $S_{y_i} \leftarrow S_{y_i} \cup \{s_i\}$ ;
17: end for
18:  $D_f \leftarrow GMM(S_{y_i}, \zeta)$  while  $k \in 1, \dots, K$ ;
19:  $D_c \leftarrow D_i \cap D_f$ ;
20:  $D_e \leftarrow D_i \cup D_f - D_c$ ;
21:  $D_h \leftarrow D - D_i \cup D_f$ ;

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When employing the label selection, we ensure stability by conducting two random data augmentations per group of views. Inspired by PES [17], we average the predicted confidences as selection criteria subsequently. To select the data based on features, we reference FINE [18] to first construct the gram matrix for each class k ($k \in [1, K]$ and K is the number of classes in the dataset), i.e., after traversing each set of data $\{x_i^j | i \in N, j \in M\}$ to obtain the predicted feature f_i , we update the gram matrix for the labeled class y_i ($y_i \in [1, K]$). Then the gram matrix is eigen decomposed and the generated principal component E_k is used as the class-representation vector of the k th class. Then we put the data $\{x_i^j\}$ in the k th class whose feature representation can be aligned with E_k into the data set D_f . This is done by computing the inner product of $\{f_i | y_i = k\}$ with E_k and fitting it to a Gaussian mixture model (GMM), and $\{x_i^j\}$ that lie in the clean clusters with probability greater than the hyperparameter ζ will be placed in D_f .

4.3.2. Division

We take the intersection of D_i and D_f to obtain the clean data set D_c , the data that satisfies only one of the above conditions as the edge data set D_e , and the last data as the hard dataset D_h . For samples in the data set D_c , the sample features align closely with the categories of the data labels, and the predicted labels match the data labels, indicating a low probability of label noise. In dataset D_e , dissimilar sample features or mismatched predicted labels suggest potential label noise. The former may be attributed to model overfitting, where it mistakenly learns noise label category features at the expense of real label category features. The latter may arise from subtle differences between noise and real label category features, making it difficult to determine if the model has accurately learned the latter. Lastly, samples in the dataset D_h exhibit both dissimilar sample features and predicted labels compared to their data labels, indicating a high likelihood of noisy labels. Therefore, we can consider the credibility of the decreases in the sequence of D_c , D_e , and D_h .

According to the convention of existing sample selection methods, the untrustworthy sample set D_h will be discarded after selection, and D_c and D_e are used for subsequent training. However, the information such as the noise ratio of the train data set is unknown, we cannot accurately determine the timing of sample selection, which will affect the effectiveness of the data selection mechanism. And there are no perfect selection criteria, so it can never achieve the precision of 100% selection. Therefore, considering that there is still some accurate data in D_h , we decide to utilize D_c , D_e together with D_h for subsequent training.

4.3.3. Training loss

Based on the above selection condition, we can assume that D_c , D_e , and D_h have more data on the noise of the label in order. Therefore, referring to DISC [34], we adopt different regularization methods to calculate the losses of the three data sets to increase the precision of recognition and improve the generalization ability of DNN.

For clean data sets D_c , we use traditional cross entropy (CE) loss L_c to avoid potential issues such as performance drop off, overfitting, and data feature neglect that can result from robust loss functions:

$$L_c = -\frac{1}{N_c} \sum_{i=1}^{N_c} \log(\text{pred}_i((x_i^1, \dots, x_i^M), y_i)) \quad (2)$$

For edge data set D_e , we need to consider the possible noise label problem in it, so the selected loss function should be more robust. Ghosh et al. [30] suggest theoretically that the mean absolute error (MAE) can be robust to noisy labels under certain assumptions. However, the robustness of MAE can simultaneously lead to increased training difficulty and degraded performance. In contrast, CE loss is easy to optimize but sensitive to label noise. Therefore, we choose the generalized cross-entropy (GCE) loss [19], which uses hyperparameter q to control the relative importance of MAE and CE in GCE loss. From a derivative point of view, GCE is equivalent to CE when q is close to 0, and when q is close to 1, GCE is equivalent to MAE. Thus, the advantages of MAE and CE are combined to calculate the loss value L_e more robustly, and the probability problem of the existence of noisy labels is considered at the same time:

$$L_e = -\frac{1}{N_e} \sum_{i=1}^{N_e} \left(\frac{1 - \text{pred}_i((x_i^1, \dots, x_i^M), y_i)^q}{q} \right) \quad (3)$$

For the hard dataset D_h , different from D_e , we need to find a small amount of valuable clean data from a large amount of label noise data instead of excluding possible label noise data from a large amount of clean data. To utilize D_h that contains the most noise data from labels, we use the noise-resistant data augmentation technique Mix [20] applied to the union D_m of the three subsets, ie $D_m = D_c \cup D_e \cup D_h$. The Mixup can be denoted as $(\tilde{x}_i^1, \dots, \tilde{x}_i^M) = \lambda(x_i^1, \dots, x_i^M) + (1 - \lambda)(x_{b(i)}^1, \dots, x_{b(i)}^M)$, $\tilde{y}_i = \lambda y_i + (1 - \lambda)y_{b(i)}$, $\lambda \sim \text{Beta}(\alpha, \alpha)$, where $(x_i^1, \dots, x_i^M)$ and $\tilde{y}_i$ are linearly interpolated views and label between index i and another random index $b(i)$ in the same mini-batch of D_m . Mixup's loss function provides a regularizing effect to the model by mixing samples and corresponding weighted loss. This approach encourages the model to be more robust to small variations in input samples, as it sees more samples of intermediate states during training. And Mixup reduces the model's dependence on specific training samples by increasing the diversity of training samples, thereby reducing the risk of overfitting. During the training process, the D_m is included with a binary cross-entropy loss (BCE):

$$L_{bce}((x_i^1, \dots, x_i^M), y_i) = - \sum_{k=1}^K (y_{ik} \log(\text{pred}_i(k)) + (1 - y_{ik}) \log(1 - \text{pred}_i(k))) \quad (4)$$

$$L_m = \frac{1}{N} \sum_{i=1}^N L_{bce}((\tilde{x}_i^1, \dots, \tilde{x}_i^M), \tilde{y}_i) \quad (5)$$

Then the overall training loss L can be expressed as Eq. (6), where λ_1 and λ_2 are the hyperparameter of L_e and L_m respectively to balance the loss value:

$$L = L_c + \lambda_1 * L_e + \lambda_2 * L_m \quad (6)$$

5. Experiments

In this chapter, we conducted parameter experiments based on the proposed method on 3DWebNet40, and then evaluated and verified the effectiveness of the method through comparative experiments with state-of-the-art methods and ablation experiments.

5.1. Experimental setups

The following experiments are all performed on the presented 3DWebNet40 dataset illustrated in Section 3. The overall process of the experiment can be expressed as follows: First, we use the early stopping mechanism to train w epochs and obtain the feature extractor g and the classifier c . Then we perform joint sample selection and utilize the obtained three subsets for the training of the next m epochs, i.e., selecting data once every m epochs until the end of training. The early stopping mechanism refers to training CNN with fewer epochs to achieve an under-fitting or just-fitting state, making it easier to filter

Table 3
Accuracy (%) when λ_1 and λ_2 take different values

λ_1	λ_2			
	0.2	0.5	0.8	1.0
0.2	93.99	93.99	94.13	93.19
0.5	94.16	94.23	94.28	94.23
0.8	94.13	94.25	94.11	94.28
1.0	93.99	94.02	94.13	94.28

out label noise. In addition, to ensure the effective performance of the method, we add a single-view pre-training process before multi-view learning.

In the following experiments, we choose Resnet18 as the backbone and we render a set of 2D images with a size of 224×224 for each 3D shape. Then, each channel of the rendered grayscale view is set to the same value, which is converted into an RGB image for network training. To save time, we randomly select 20 classes among the 40 classes for the parameter experiment and use the data of all 40 classes for comparison experiments and ablation experiments.

In the parameter experiment, we pre-train the single view network for 2 epochs and train the multiview network for 50 epochs, where the warm-up training epoch w is 2. The learning rate is set to 5×10^{-5} , the batch size is 6, and the learning rate drops to 2.5×10^{-6} at the 30th epoch. In other experiments, the single-view pretraining is 5 epochs, and the epoch of multiview network training is set to 60. Sample selection starts after the 5th epoch and batch size is 20. The learning rate is set to $5e-4$ and drops to 50% of the previous value every 30 epochs. In all the experiments, we use SGD as the optimizer, the momentum and weight decay are 0.9 and 0.001 respectively, and m is set to 10.

5.2. Hyperparameter setting

There are three hyperparameters in our method, which are the feature confidence ζ in the joint sample selection stage, the λ_1 , and λ_2 used to balance the three kinds of loss. Referring to FINE [18], we fix ζ to 0.5 empirically. We set different values to compare the effect of the coefficient λ_1 of L_e and λ_2 of L_m on the training results, and the experimental results are shown in Table 3. It can be seen that the same accuracy can be achieved for different values. And although the training effect generally increases with the increase of λ_1 and λ_2 , the difference between the minimum (93.19%) and maximum (94.28%) values indicates that the model is relatively insensitive to these two hyperparameters. In addition, for the correlation between λ_1 and λ_2 , it can be found that as λ_1 (λ_2) increases, the value of λ_2 (λ_1) to achieve the best effect also increases, that is, the gap between $\lambda_1 = 0.8$ and $\lambda_1 = 1.0$ ($\lambda_2 = 0.8$ and $\lambda_2 = 1.0$) gradually decreases as λ_2 (λ_1) increases.

Based on the experimental data, we select one of the highest accuracy results to set λ_1 at 1 and λ_2 to 1 for the next step of the experimental analysis. And according to the above analysis, we have reason to believe that this setting will bring the best recognition effect.

5.3. Comparison with the state-of-the-art

To validate the effectiveness of our proposed approach, we conducted comparative experiments with some typical state-of-the-art methods. We show the performance of several representative view-based methods, such as MVCNN [10], GVCNN [37], MHBN [23], MVT [38], DAN [39] and MVDAN [40].

Table 4 displays the identification results of all methods compared. The results show that our method achieves the best result on 3DWebNet40 with a recognition accuracy of 90.04%. It is about 2% higher than the second-best method DAN and 4.4% higher than the baseline method MVCNN, demonstrating the effectiveness of our processing of background noise and label noise data. The worst performance is

Table 4

Comparison of the Top-1 accuracy (%) of 3D shape recognition experiments on 3DWebNet40.

Method	Accuracy
MVCNN	85.64
GVCNN	87.67
MHBN	78.55
MVT	68.87
DAN	88.08
MVDAN	87.20
Ours	90.04

Table 5

Comparison of the Top-1 accuracy (%) of 3D shape recognition experiments on ModelNet40.

Method	Label noise ratio		
	0.0	0.1	0.2
MVCNN	94.94	92.50	87.12
GVCNN	95.54	90.48	89.51
MHBN	88.37	85.94	85.41
MVT	87.07	82.58	75.45
DAN	93.48	89.67	88.98
Ours	95.83	92.59	90.11

MVT with an accuracy of only 68.87% and we guess this is caused by the lack of pre-training on single views. In short, the improvement in recognition accuracy proves the effectiveness of our method in 3D shape recognition tasks. Fig. 6 shows the classification accuracy of our model for each specific category. From this we can see that the most confusing categories are flower pot and plant, probably because the two tend to go together.

In addition to this, we also conduct experiments on the standard dataset ModelNet40 with different label noise levels added. Table 5 shows the classification performance of different models in ModelNet40 with noise ratios of labels of 0.0, 0.1 and 0.2, respectively. According to the experimental results, our method performs well under different label noise ratios. And as the ratio of noisy labels increases, the recognition accuracy of the proposed method decreases the least compared to other models. This shows that our proposed 3D model classification method based on regional attention mechanism and joint sample selection not only improves the performance of 3D model classification under the condition of no label noise, but also effectively alleviates the problem of recognition accuracy degradation caused by the increase of label noise.

5.4. The effect of the region attention module

To more effectively reduce the impact of background noise, we intend to superimpose the region attention module on different levels of the ResNet18 backbone network. Taking into account factors such as algorithm accuracy and time performance, we consider three placement ways, P_1 : at the end of each basic block, P_2 : at the end of the second basic block of each Stage, and P_3 : at the end of each basic block of the last Stage. We also evaluate the method without the region attention module called P_4 . The specific comparison results on 3DWebNet40 are shown in Table 6.

We compare the above placement ways from three aspects: recognition accuracy, prediction time of each batch, and model size. We can see that P_2 can achieve the best results in all aspects, which is about 0.6% higher than the model without the region attention module.

According to the experimental results, when P_2 is placed (i.e. at the end of the second basic block in each Stage), the best recognition accuracy (90.04%) and recognition speed (0.143 s) are achieved. By analyzing the similarities and differences between the four placement methods, it can be seen that P_1 and P_3 add regional attention modules at the end of two basic blocks in one or more stages, and the effect

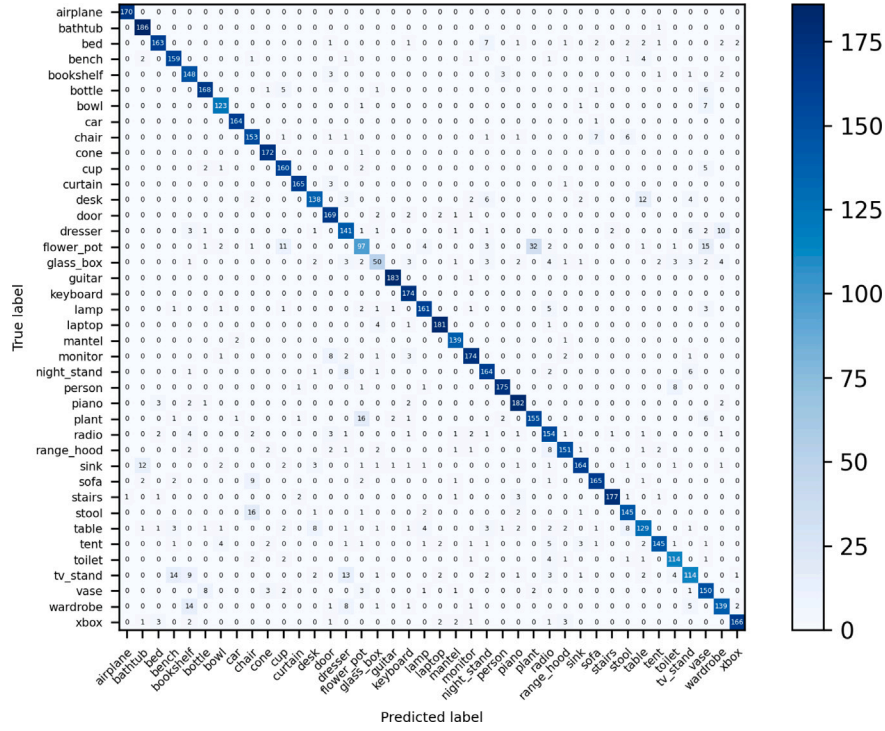


Fig. 6. Our model confusion matrix on 3DModelNet40.

Table 6
Effect of region attention module.

	P_1	P_2	P_3	P_4
Acc/%	89.31	90.04	89.89	89.41
Time/s	0.230	0.143	0.231	0.229
Size/M	11.20	11.20	11.20	11.20

is roughly the same as that of P_4 without adding regional attention modules. However, P_2 only adds the regional attention module at the end of the second basic block of multiple stages, but significantly improves the recognition effect and speed of the model. This indicates that adding the regional attention module after the first basic block of each Stage of ResNet18 may reduce the recognition effect and performance of the model, while adding the regional attention module after the second basic block of each Stage will improve the recognition effect and performance of the model. Moreover, the recognition effect and performance of the model are slightly improved than decreased.

5.5. The effectiveness of joint sample selection

We implemented ablation experiments to demonstrate the validity of sample selection, as shown in Table 7.

According to the analysis of experimental results, we can see that our proposed method (90.04%) is better than the method of single selection criteria (89.39% and 89.86%), which proves the effectiveness of our joint sample selection mechanism. Compared with the union of data filtered by two criteria, our proposed method can achieve higher recognition accuracy by dividing different data sets and applying different regularizations for training. In addition, the performances of using the Mixup strategy or discarding D_h alone are all worse than the combination of the three subsets, proving that the rationalization of dataset partitioning. And the worst recognition effect (88.78%) is achieved when D_h is discarded, which verifies our conjecture that there are still some clean data in D_h proposed above. In short, the three divided subsets that use the corresponding regularization computations all play an indispensable role in the joint sample selection.

Table 7
Different sample selection criteria.

Select Criteria	Accuracy (%)
D	89.49
D_m	89.38
D_l	89.39
D_f	89.86
$D_l \cup D_f$	89.86
$D_c + D_e$	88.78
$D_c + D_e + D_m$	90.04

D : Use the original dataset.

D_m : Apply Mixup on the original dataset.

D_l : Only label selection.

D_f : Only feature selection.

$D_l \cup D_f$: Take the union of the results of the above two selection criteria.

$D_c + D_e$: Use different regularization methods to train D_c and D_e divided after joint selection.

$D_c + D_e + D_m$: Use different regularization methods to train D_c , D_e and D_m obtained after joint selection.

6. Conclusion

In this paper, we investigate the problem of webly supervised 3D shape recognition. Specifically, we first construct a benchmark webly supervised 3D shape dataset 3DWebNet40 which contains a total of 36,665 shapes from the 40 semantic concepts of the ModelNet benchmark. Then we propose an effective learning method to address both label noise and background noise. We employ a region-attention module to focus on the target object to mitigate the influence of background noise. Meanwhile, to address label noise, we propose a joint sample selection mechanism that uses both label selection and feature selection to get three divided subsets. Then we employ different loss calculations separately to make each subset fully functional, reducing the damage from the label noise. Extensive experiments show that our approach achieves state-of-the-art performance compared to existing work. Currently, we use a fixed camera setting to capture the views, which might not obtain the complete information of the target 3D shape. Our future work will be to extend our framework to support

the dynamic viewpoint setting. In this paper, joint sample screening is used to weaken the impact of 3D label noise, which provides a new thread for researchers to consider combining more dimensional sample selection criteria to further suppress the label noise.

CRedit authorship contribution statement

Xizhong Yang: Validation, Investigation, Formal analysis, Data curation., **Qi Guo:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Data curation, Conceptualization., **Wenbin Chen:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization., **Mofei Song:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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